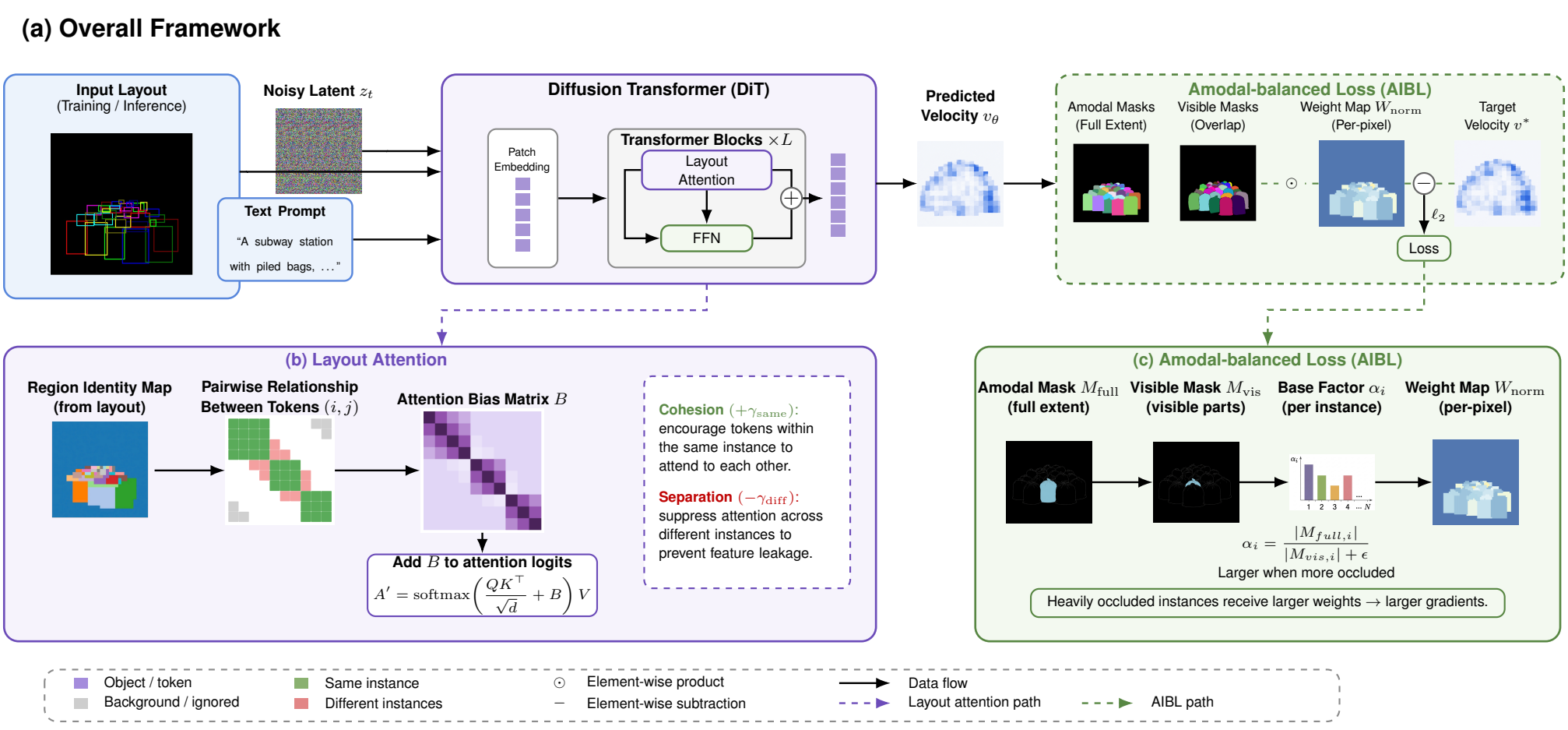

(a) Overall Framework
Input Layout
(Training / Inference)
Noisy Latent z_t
Text Prompt
"A subway station with piled bags, ..."
Diffusion Transformer (DiT)
Patch Embedding
Transformer Blocks $\times L$
Layout Attention
FFN
Predicted Velocity v_θ
Amodal-balanced Loss (AIBL)
Amodal Masks (Full Extent)
Visible Masks (Overlap)
Weight Map W_{norm} (Per-pixel)
Target Velocity v^*
ℓ_2
Loss
(b) Layout Attention
Region Identity Map (from layout)
Pairwise Relationship Between Tokens (i, j)
Attention Bias Matrix B
Add B to attention logits
$A' = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}} + B\right) V$
Cohesion $(+\gamma_{\text{same}})$: encourage tokens within the same instance to attend to each other.
Separation $(-\gamma_{\text{diff}})$: suppress attention across different instances to prevent feature leakage.
(c) Amodal-balanced Loss (AIBL)
Amodal Mask M_{full} (full extent)
Visible Mask M_{vis} (visible parts)
Base Factor α_i (per instance)
Weight Map W_{norm} (per-pixel)
$\alpha_i = \frac{|M_{full,i}|}{|M_{vis,i}| + \epsilon}$
Larger when more occluded
Heavily occluded instances receive larger weights → larger gradients.
Object / token
Background / ignored
Same instance
Different instances
Element-wise product
Element-wise subtraction
Data flow
Layout attention path
AIBL path